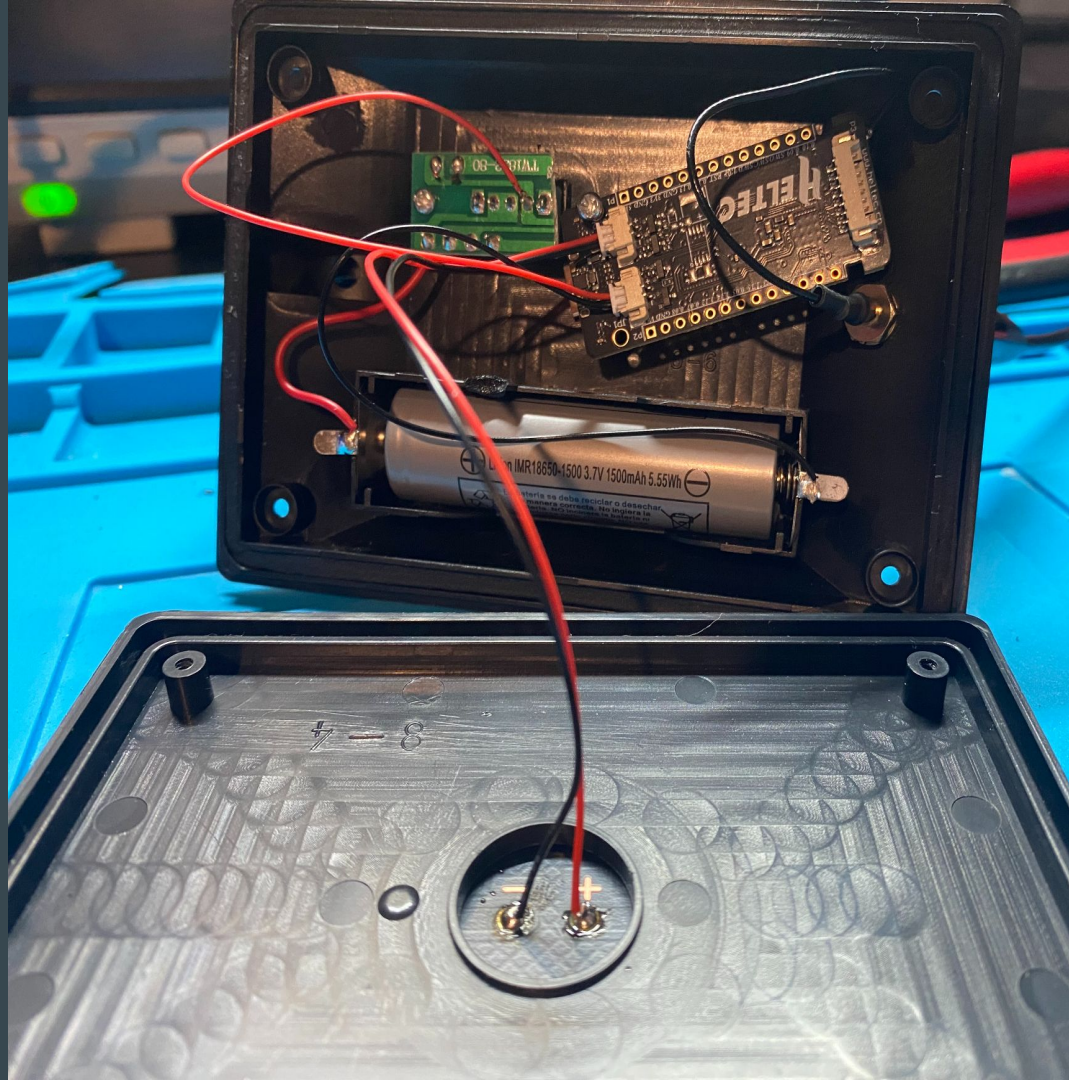


meshtastic essentials

prepared by

Danielle McGuire (*Protomesh Collective*)
and **Greyfox** (*Hacker's Guild*)
for **Art&&Code: Transmissions**

2026 Sep 20



How much do you already know?

- 0 "Mesh-what?"
- 1 Heard the name
- 2 Read about it
- 3 Own a device
- 4 Sending messages
- 5 Building solar nodes

SLIDE SHAMELESSLY STOLEN FROM SUNBEAR

what do you think Meshtastic is?

shout it out! no wrong answers

SLIDE SIMILARLY STOLEN FROM SUNBEAR

Today We Will:

DISCUSS

1. Meshtastic use cases
2. Meshtastic node hardware
3. Meshtastic node firmware
4. Meshtastic client software
5. Meshtastic configuration
6. Meshtastic sensors

DO

1. Identify node components
2. Assemble nodes
3. Flash Meshtastic firmware onto nodes
4. Connect to nodes via Bluetooth app
5. Configure nodes with ideal settings for Pittsburgh
6. Exchange messages and view sensor data

PART 1 OF 6: INTRODUCTION

Discuss: Meshtastic Use Cases (Dani)

Do: Identify Meshtastic components
(Greyfox)

Danielle McGuire (she/her)

Industrial Controls Systems (ICS) Cybersecurity Practitioner

2016-present

Industry Advisor, Pitt Cyber Energy Center

2024-present

Founder/Secretary, Society of Hackers

Sep 2025-present

Founder/2nd Coordinator/1st Scientist, Protomesh Collective

Feb 2026-present

likes: history, gardening, cooking, hacking

dislikes: oppression, bigotry, *Harry Potter*

Come see me keynote the October 2026 session of the Society of Hackers with
*Darkness Within Darkness: A Fun and Practical Introduction to Quantum
Teleportation*

Grey Fox

Former Air Force (retired)

DAY: Pentesting / Vulnerability Hunting / Tolerating AI

NIGHT: Digital Privacy / Physical Security / Hating AI

WEEKENDS: Combat tactics and casualty care

ALWAYS Obsessed with wireless tech and anti-surveillance

Rediscovering my goth/metal head wardrobe

PART 1 OF 6: INTRODUCTION

Discuss: Meshtastic Use Cases (Dani)

Do: Identify Meshtastic components
(Grey Fox)

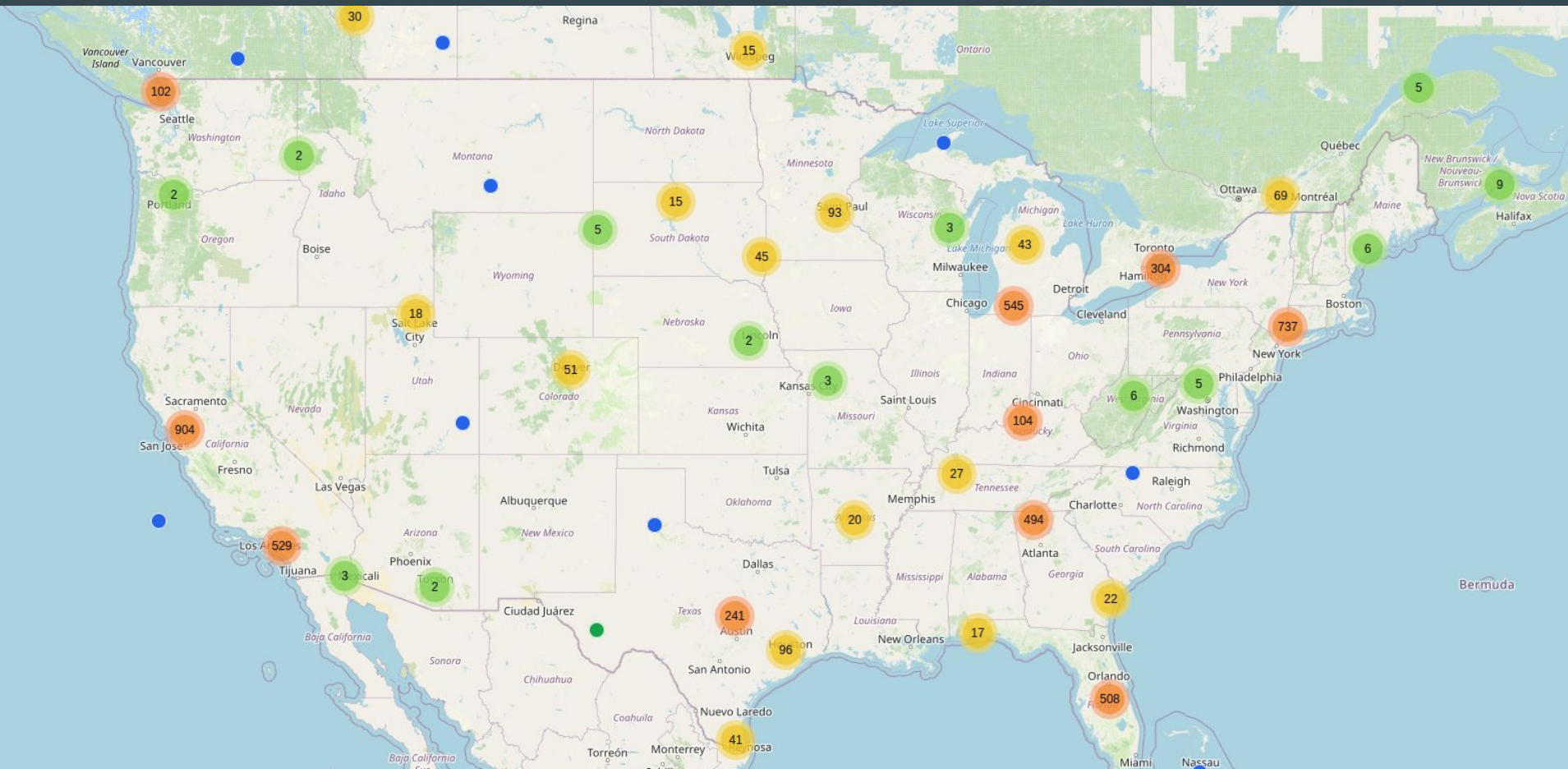
What is Meshtastic?

- Free and open source communications technology
 - Text messaging over radio
 - Decentralized, encrypted, and hyperresilient
- The nodes themselves are the network
 - Messages hop from node to node
 - More nodes increase network coverage
- Uses LoRa (“LOng RAnge”) radio waves
 - 900-930 MegaHertz band centered on 915 MHz (33cm wavelength)
 - ½ to 1 mile of range with stock antenna

Why Meshtastic?

- Hyperresilient - it works when nothing else does
 - Doesn't need cell signal, Internet, or even the power grid
 - Ideal for disaster recovery AND remote sensor deployments
- Decentralized and community-driven
 - Don't need to wait for or pay for centralized network infrastructure
 - Volunteer/open-source projects bring out the best of people and their skills
 - "Pittsburgh has the best Mesh we've seen" - Anthrocon 2026
- End-to-end encrypted
 - Meshtastic's cryptosystem is NOT as good as Signal's!
 - But, Signal gets to assume a reliable network, and ham radio doesn't allow encryption at all
 - Encryption is convivial technology that enables experimental projects to become practical comms

United States Coverage Map (Feb 2026)



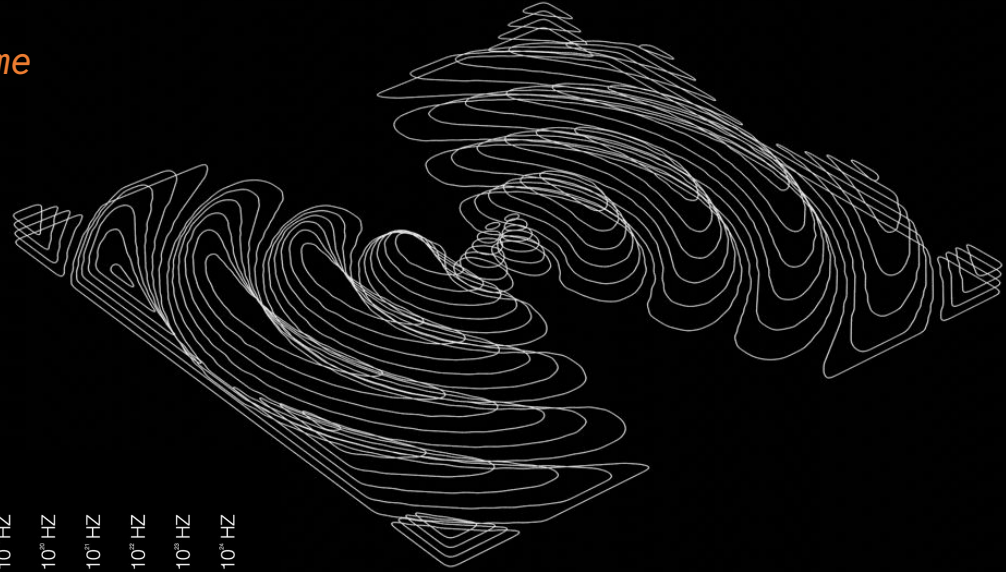
Q



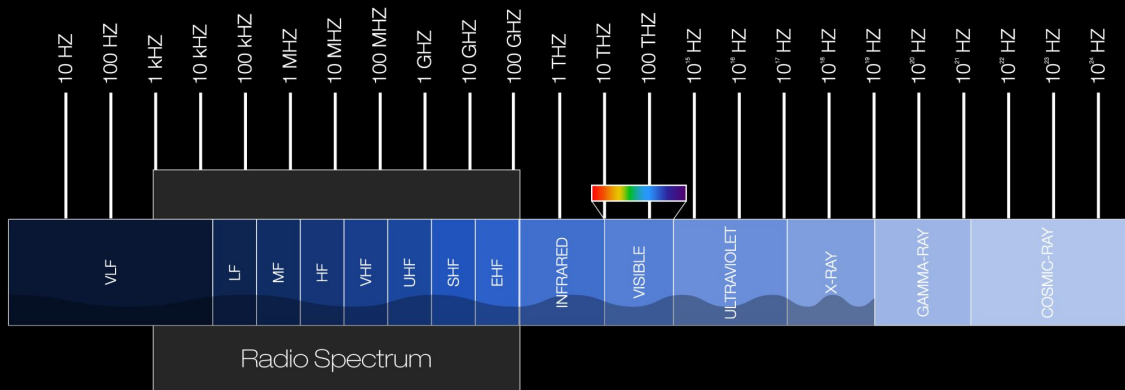
"RADIO, OH RADIO..."

...do you believe there's still some magic left somewhere inside our souls?"

"Howl", The Gaslight Anthem



Animation of radio waves from dipole antenna rippling through 3D space (<https://blender.stackexchange.com/questions/304616/creating-an-animation-of-radio-waves>)



radio 101

a radio antenna **tuned** to the right frequency can receive the electromagnetic wave and convert it into **electric current**

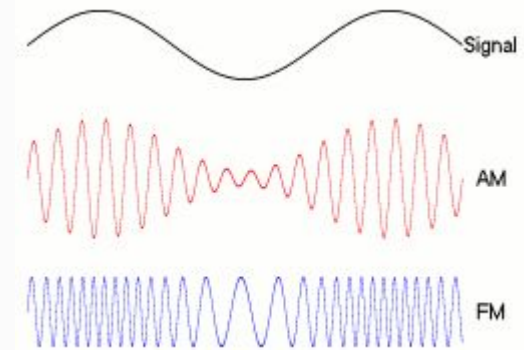
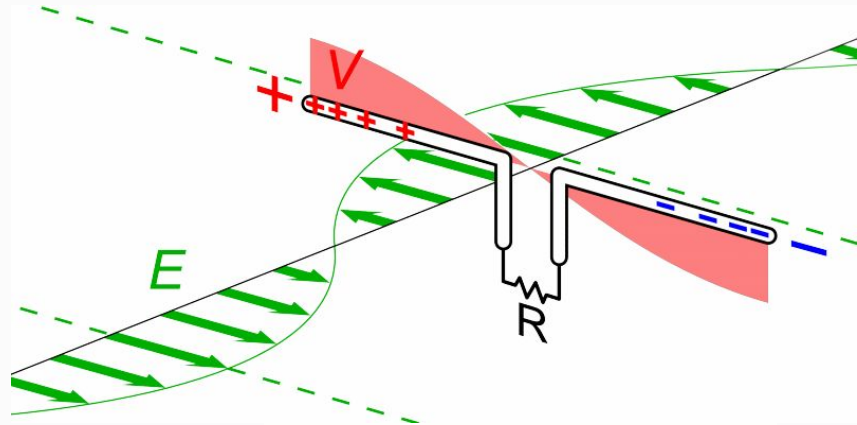
a radio **receiver (RX)** is a circuit that decodes the information encoded in the received wave, and processes/outputs that information

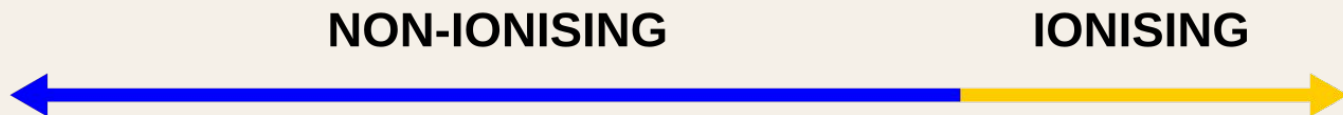
some ways to encode information in a **carrier wave**:

Amplitude Modulation (AM)

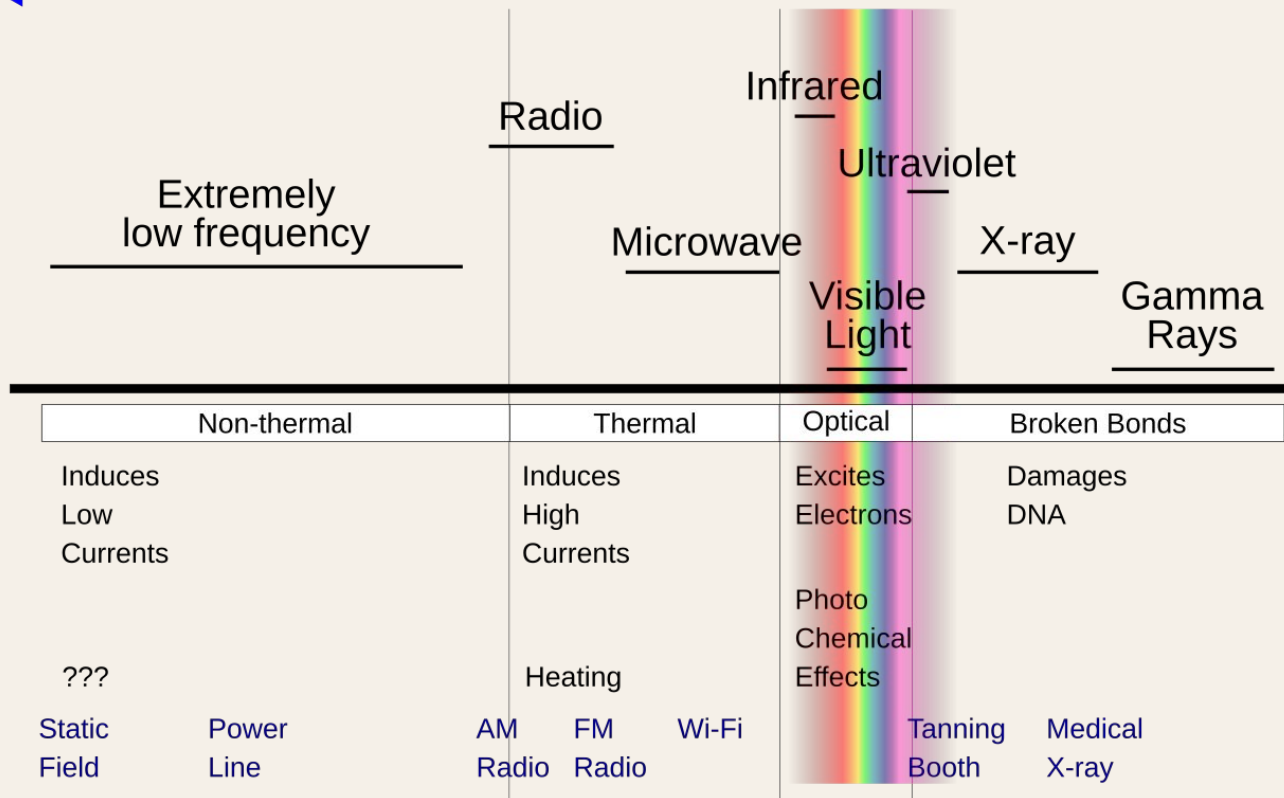
Frequency Modulation (FM)

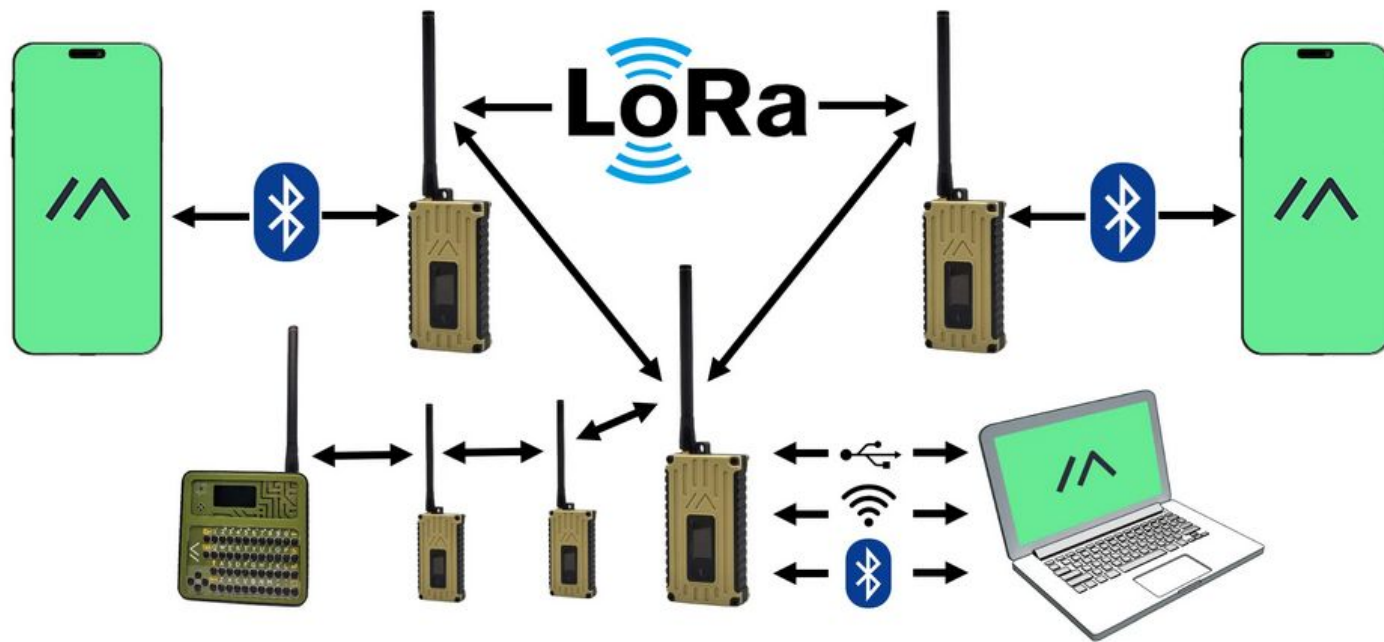
LoRa (too complex to draw easily, but it's just another way to encode information!)





The Electromagnetic Spectrum (from Wikipedia)





How does the Meshtastic network work?

(US) Meshtastic nodes talk to each other over (915 MHz) LoRa radio waves. We usually talk to nodes via Bluetooth, but we can also use USB or sometimes Wifi.

How do Meshtastic messages travel?

1. Your phone pairs to the device over bluetooth
2. You send your text message to the device
3. The device sends the message to the network over LoRa
4. The nodes in the network repeat the message a certain number of hops, hopefully reaching the recipient
5. Their device sends the message over Bluetooth to their phone



Real-World Examples

- Asheville, North Carolina
 - After Hurricane Helene destroyed large chunks of Asheville's infrastructure, people were able to use combinations of Meshtastic and GMRS radios to spread information when the cell towers were out. There are still 65 active nodes in the Asheville portion of the North Carolina Mesh Network.
- Redtail Resilience Network
 - Enabling both hyperlocal community formation in Pittsburgh and disaster-recovery volunteer coordination from East Palestine to New Orleans
 - Led in part by Sunbear, who also helps lead WPAmesh
- Ukraine
 - Meshtastic is being used for both combat and non-combat roles where access to encrypted radios is limited. Meshtastic pairs well with ATAK systems that allow soldiers to have real-time maps

PART 2 OF 6: BUILDING NODES

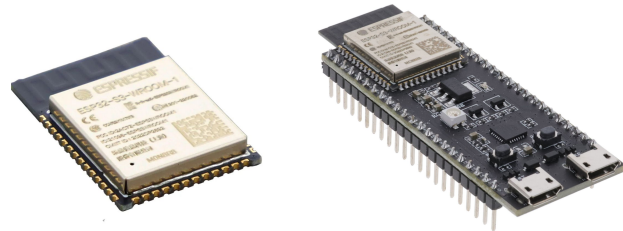
Discuss: LoRa Node Hardware (Dani)

Do: Assemble nodes (Greyfox)

Chipsets - Interfaces for LoRa Radios

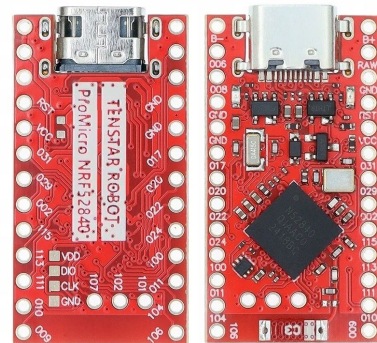
ESP32-S3 ("ESP32 chipset")

High performance and processing
BLE + WiFi (supports MQTT)
I/O device and display support
Ideal for mobile handheld nodes



nRF52840 ("Nordic chipset")

Power efficiency
BLE only
Highly stable
Ideal for solar repeater nodes



Boards - Complete Hardware Packages

- Lilygo: T-Echo, T-Deck, T-Beam, LoRa T3-S3
- RAK Wireless: WisMesh, WisBlock RAK4631
- Seeed: Card Tracker, Wio Tracker
- Heltec: T114, V4
 - Protomesh Collective: HACKERSBLOOM, NOMAD



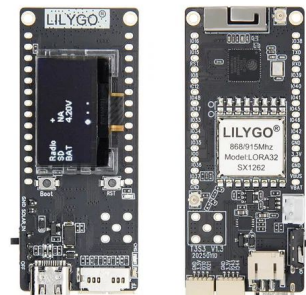
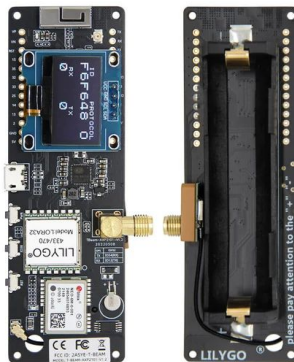
ESP32



nRF52840

Boards - Lilygo

Lilygo: T-Echo (\$63 Amazon), T-Deck (\$60), T-Beam (\$55),
LoRa T3-S3 (\$22)



Boards - RAK Wireless

RAK Wireless: WisMesh (\$90 vendor pre-shipping), WisBlock RAK4631 (\$18), WisBlock Starter Kit (4631+19007 bundle, \$25)



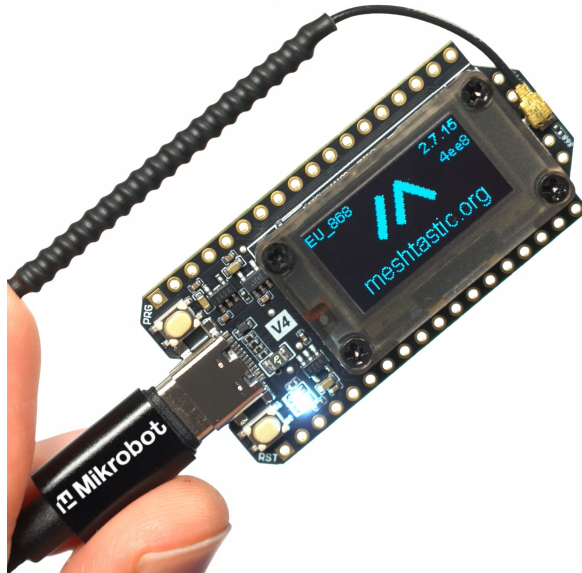
Boards - Seeed

Seeed: SenseCAP Card Tracker T1000-E (\$40 vendor pre-shipping),
Wio Tracker (\$53)



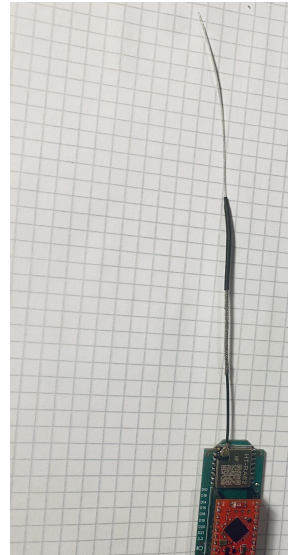
Boards - Heltec

Heltec: T114 (\$18 from vendor pre-shipping), V4 (\$20), HT-RA62 (bare LoRa radio, \$4)



Boards - Protomesh Collective

Protomesh Collective: HACKERSBLOOM (\$30 NOTAFLOF), NOMAD (\$30 NOTAFLOF), (also, please enjoy these unrelated pictures of a hand-soldered “FakeTec V4” (\$9 cost) w/ DIY “Steven’s Stinger V3” antenna (\$1 cost), and the prototype KILN Solder Reflow Oven about to enter public beta at Prototype x Oakland)

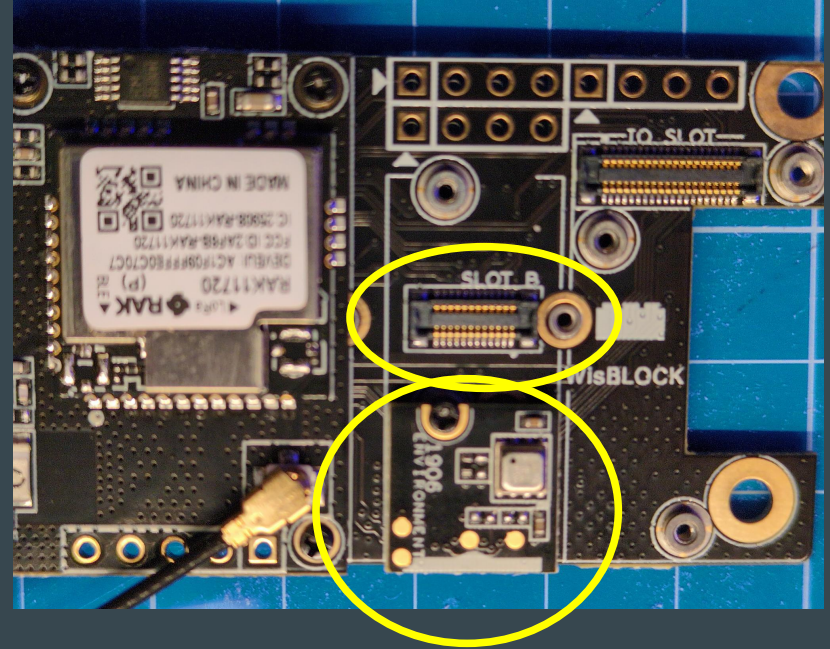
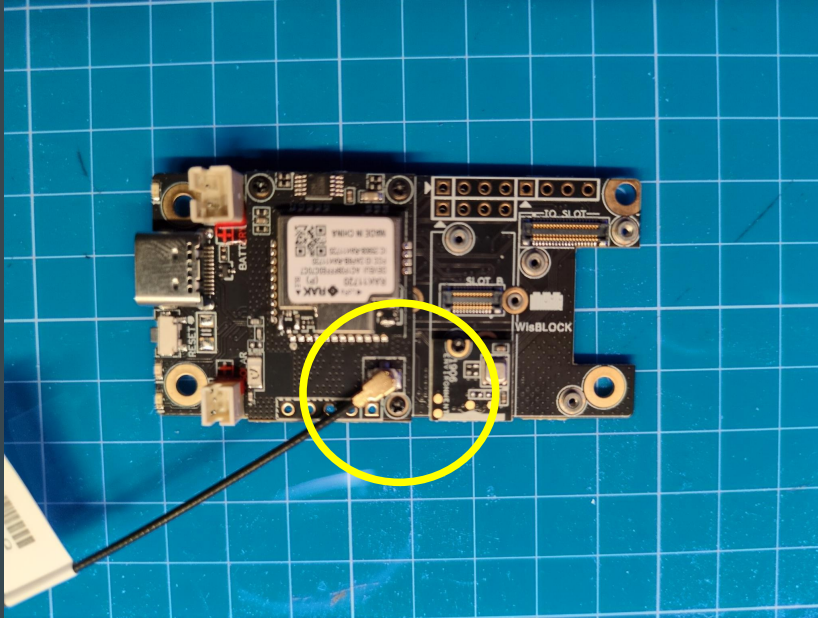


WisBlock RAK4631

- Ready-made for sensors
 - Snap-on other RAK sensors (1901, 1906, etc.)
 - Smallest screw for securing
 - Auto-detected by app
- Never forget the antenna
 - Meshtastic firmware transmits (TX) on power-up
 - TX without an antenna traps electrons
 - This is bad
 - Snap-on (sort of) to IPEX port



Assembly



NOTE: This is a RAK11722, not a RAK4631, but the base is still RAK19007. Layout is the same for sensor seats and IPEX port

Reading Telemetry

- Telemetry & Detection Sensor
 - Telemetry controls environmental and device status readings
 - Detection is passive IR for movement, magnetic sensors, etc.
- Hardware is small and modestly priced
 - Sensor readings pushed to mesh at preset intervals
 - Your WisBlock allows snap-on sensor chips
- Device ROLE as Sensor or Tracker
 - Dedicated IoT deployments
 - Sleep mode until preset broadcast interval
 - In either role, normal messaging is disabled

PART 3 OF 6: FLASHING NODES

Discuss: Meshtastic Firmware (GFX)

Do: Flash Meshtastic onto nodes
(Dani)

Meshtastic Web Flasher



<https://flasher.meshtastic.org/>

The screenshot shows the Meshtastic Web Flasher interface. At the top is a green double-slash logo. Below it, the text "MESHTASTIC WEB FLASHER" is displayed in a green, pixelated font. A subtitle reads: "Update your Meshtastic device with official firmware, straight from your browser". The interface is divided into three main steps: 1. DEVICE, 2. FIRMWARE, and 3. FLASH. Each step has a corresponding icon and a button. Step 1 shows a device icon and a "Select Target Device" button. Step 2 shows a folder icon with a download arrow and a "Select Firmware" button. Step 3 shows a lightning bolt icon and a "Flash" button. Below each step is a brief instruction. At the bottom, there are links for "Open Serial Monitor", "Meshtastic Docs", "Contribute on GitHub", and a "Language" dropdown menu.

MESHTASTIC WEB FLASHER

Update your Meshtastic device with official firmware, straight from your browser

- 1**

Select Target Device
DEVICE
Plug in your device via USB. Please ensure the cable is not a power-only one.
- 2**

Select Firmware
FIRMWARE
Choose from the release options or upload a release zip downloaded from Github.
- 3**

Flash
FLASH
Flash your device. Choose whether you wish to update your device or wipe the flash and install from scratch.

Open Serial Monitor >... Meshtastic Docs Contribute on GitHub Language

(requires Chromium-based browser)

Selecting Hardware Version



Select your connected device



All Devices

RAK

B&Q

LilyGo

Seeed

Heltec

Elecrow

M5Stack

NomadStar

muzi WORKS

esp32

nrf52840

esp32-s3

rp2040

esp32-c3

esp32-c6

ⓘ If your connected device already has Meshtastic installed, you can automatically detect it:



Auto-detect

Supported Devices

LILYGO T-Echo



nrf52840

LilyGo



RAK WisBlock 4631



nrf52840

RAK



LILYGO T-Beam Supreme



esp32s3

LilyGo



LILYGO T-LoRa T3-S3



esp32s3

LilyGo



LILYGO T-LoRa T3-S3 E-Ink

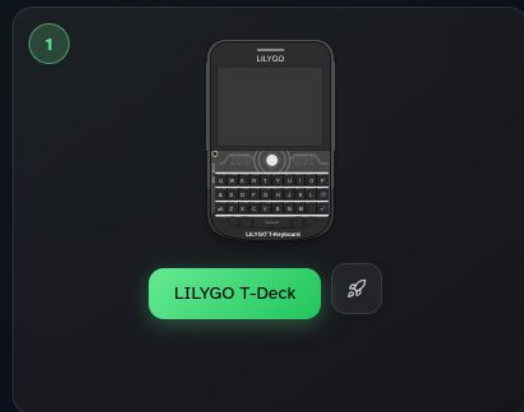


esp32s3

LilyGo

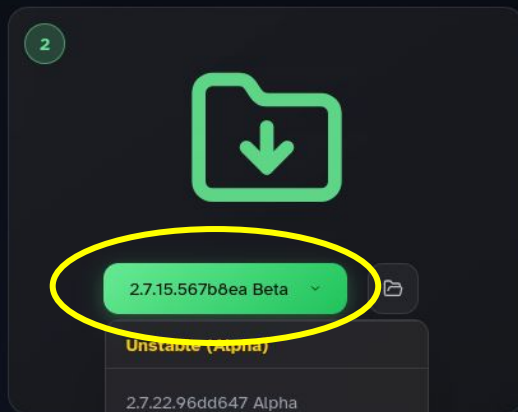


Selecting Firmware Version



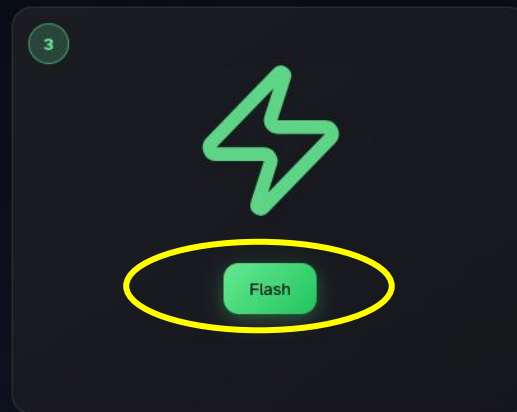
DEVICE

Plug in your device via USB. Please ensure the cable is not a power-only one.



FIRMWARE

Choose from the list or download the latest version.



FLASH

Flash your device. Choose whether you wish to update your device or wipe the flash and install from scratch.

Open Serial Monitor >_

Meshtastic

File

Language

Settings

Flashing a RAK Node



flasher.meshtastic.org wants to connect to a serial port

ttys30
ttys31
ttys4
ttys5
ttys6
ttys7
ttys8
ttys9
USB Serial (ttyUSB0)

Cancel

Connect



Flash RAK WisBlock 4631



For firmware versions < 2.2.17, trigger DFU mode manually by double-clicking RST button.



Enter DFU Mode

2

Ensure device DFU mode drive is mounted

The drive may have a different name depending on your device hardware and its bootloader.

RAK4631 (D:)

3

Download or copy UF2 file to DFU drive

Download or copy UF2 file to the DFU drive. The device will automatically reboot when the transfer completes and will start with the new firmware.



The auto reboot may cause messages about file transfer failures, write errors, or the device being ejected

Download UF2

Possible Hiccups



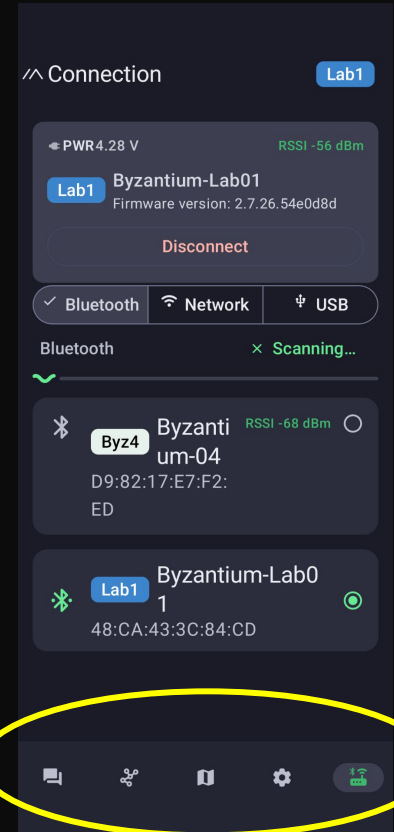
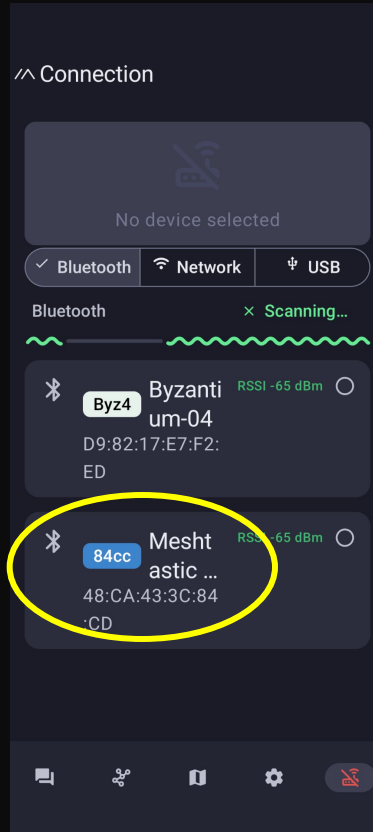
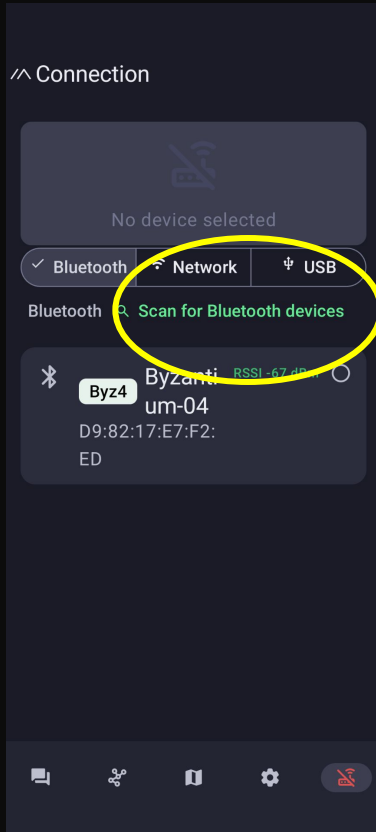
- Boot mode is not the same for all devices
- **nRF52840** usually auto-switches to bootloader, but not always
- USB cable is likely the issue
- Some GNU/Linux distros won't play ball (missing libs / dependencies)
- Windows and Mac typically work well

PART 4 OF 6: CONNECTING TO NODES

Discuss: Meshtastic Client Software
(GFX)

Do: Connect to nodes via Bluetooth
app (Dani)

Using the App (Android)



Using the App (iPhone)

Dani is herself an iPhone user.

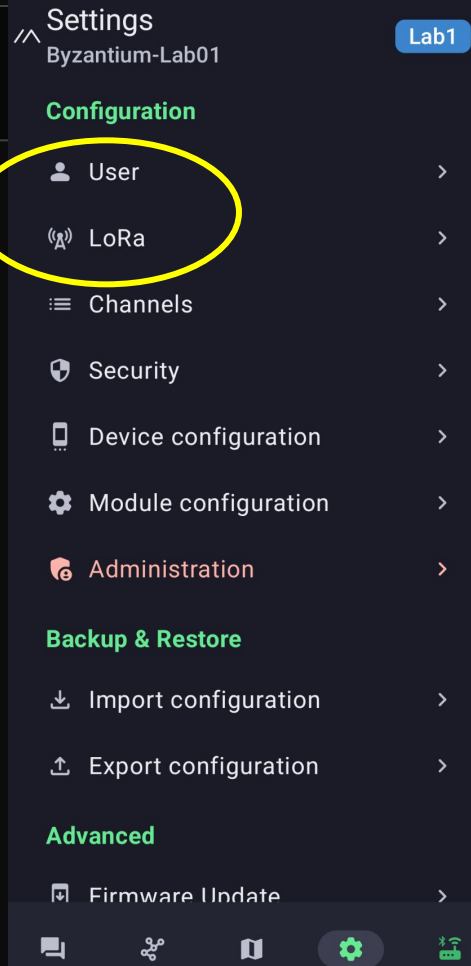
PART 5 OF 6: CONFIGURING NODES

Discuss: Meshtastic Configuration
(GFX)

Do: Configure nodes (Dani)

Baseline Configuration

- Meshtastic mobile app + BLE tether
- LoRa
- Region: United States
- Presets: LONG_FAST
- Number of Hops: 3 is default, raise based on node density
- User
 - Long Name: dealer's choice
 - Short Name: 4 characters
 - Licensed Amateur: HAM operators have a reserved space – no encryption allowed



Roles

- Device Configuration > Device
- Device Role: CLIENT
- Rebroadcast Mode: ALL

CLIENT	Default messenger. Screens, connects to app, relays if needed.
CLIENT_MUTE	Talks but never relays. Lowest power/airtime.
CLIENT_HIDDEN	Invisible in nodes list; broadcasts only when needed.
CLIENT_BASE	Home-station CLIENT; always boosts favorited nodes.
TRACKER	GPS beacon on a wake cycle; no screen.
LOST_AND_FOUND	Texts its location regularly for recovery.
SENSOR	Telemetry beacon on a wake cycle; sleeps otherwise.
TAK / TAK_TRACKER	ATAK-integration variants; the second adds tracking.
ROUTER	No BLE/WiFi, no screen; relays everything with priority.
ROUTER_LATE	Same as ROUTER, but waits for CLIENT nodes to reply first.
REPEATER	Forward-only relay; not for messaging.

Rebroadcast



Rebroadcast Mode	What it does	Best Uses / Notes
ALL (default)	Retransmits every packet it hears from the mesh, decoding each one.	Standard behavior — maximum mesh participation.
ALL_SKIP_DECODING	Rebroadcasts everything but skips decoding packets before forwarding.	Slightly faster and lower power per hop; trades off local processing of others' traffic.
LOCAL_ONLY	Only rebroadcasts packets from your local/default channel(s) — ignores foreign traffic from other channels.	Reduces airtime load in dense public meshes while staying active locally.
CORE_PORTNUMS_ONLY	Forwards only "core" packets — NodeInfo, Text, Position, Telemetry, Routing — ignoring non-standard portnums like TAK, RangeTest, PaxCounter.	Keeps infrastructure nodes from relaying experimental/optional traffic on crowded channels.
NONE	Inhibits all rebroadcasts, similar to CLIENT_MUTE behaviour.	Only permitted for SENSOR, TRACKER, and TAK_TRACKER roles — pure transmitter nodes.

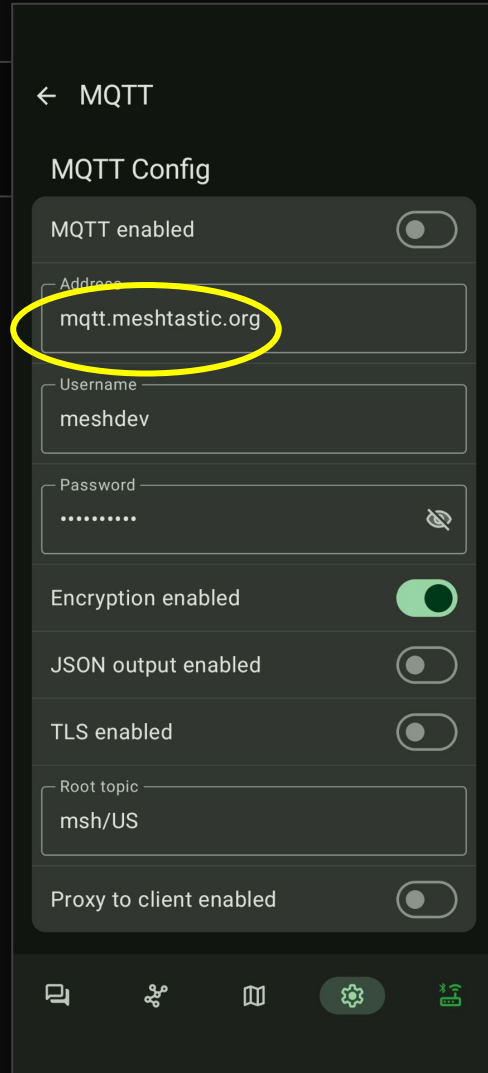
More Configuration



- **Position:** user's choice
- **Power:** important for Sensor, Tracker
- **Display:** nice-to-have options; keep default
- **Bluetooth:** no-screen devices have default PINs
 - 0000
 - 1234
 - 123456
- **Consider changing them**

Module Configuration

- MQTT: it depends
 - Default: <https://mqtt.meshtastic.org/>
 - Consider: <https://mqtt.wpamesh.net/>
- Serial: computer tethering options
- External Notification: LED, buzz, and vibrate
- Store & Forward: if acting as a server, enables infrastructure for message control
 - Only ESP32 based devices with onboard PSRAM like the T-Beam and T3-S3 can be a Store & Forward Server.



← MQTT

MQTT Config

MQTT enabled ☐

Address
mqtt.meshtastic.org

Username
meshdev

Password
..... ☐

Encryption enabled ☒

JSON output enabled ☐

TLS enabled ☐

Root topic
msh/US

Proxy to client enabled ☐

Navigation icons: Home, Search, Add, Settings, Signal strength



Sensors

- **Telemetry:** sensor + interval
- **Detection Sensor:** detection trigger type = electrical state of monitored GPIO pin
 - **LOGIC_HIGH:** high voltage rise-PIR sensors
 - **LOGIC_LOW:** Reed switch/door contact using voltage

← Telemetry

Telemetry Config

Send Device Telemetry

Enable/Disable the device telemetry module to send metrics to the mesh. These are nominal values.

Congested meshes will automatically scale to longer intervals based on number of online nodes.

Device metrics update interval

1 hour

Environment metrics module enabled

Environment metrics update interval

1 hour

Environment metrics on-screen enabled

Environment metrics use

← Detection Sensor

Detection Sensor Config

Detection Sensor enabled

Minimum broadcast (seconds)

30 seconds

State broadcast (seconds)

Unset

Send bell with alert message

Friendly name

SafeHouse01-Front

GPIO pin to monitor

Pin 0

Detection trigger type

LOGIC_HIGH

Use INPUT_PULLUP mode



Useful App Stuff



- **Nodes Tab**
 - Mark favourites
 - Check hardware
 - Send a direct message
- **Mesh Map**: only if GPS enabled
- **NodeDB Reset**: empties recorded nodes, starts over
- **Traceroute**: doesn't always work, but gives good awareness of routing and hops

Routers and Repeaters?



- Router:

- Active in path selection for packets
- Prioritised in packet route selection

- Repeater:

- Only seen/controlled via remote management
- Dumb routing, only forwarding

- Router_Late:

- Acts like a router, but let's CLIENTs rebroadcast first
- Not prioritised in routing-eases congestion

PART 6 OF 6: USING NODES

Discuss: Channels, integrating sensor modules (Dani)

Do: Exchange messages and view sensor data (Greyfox)

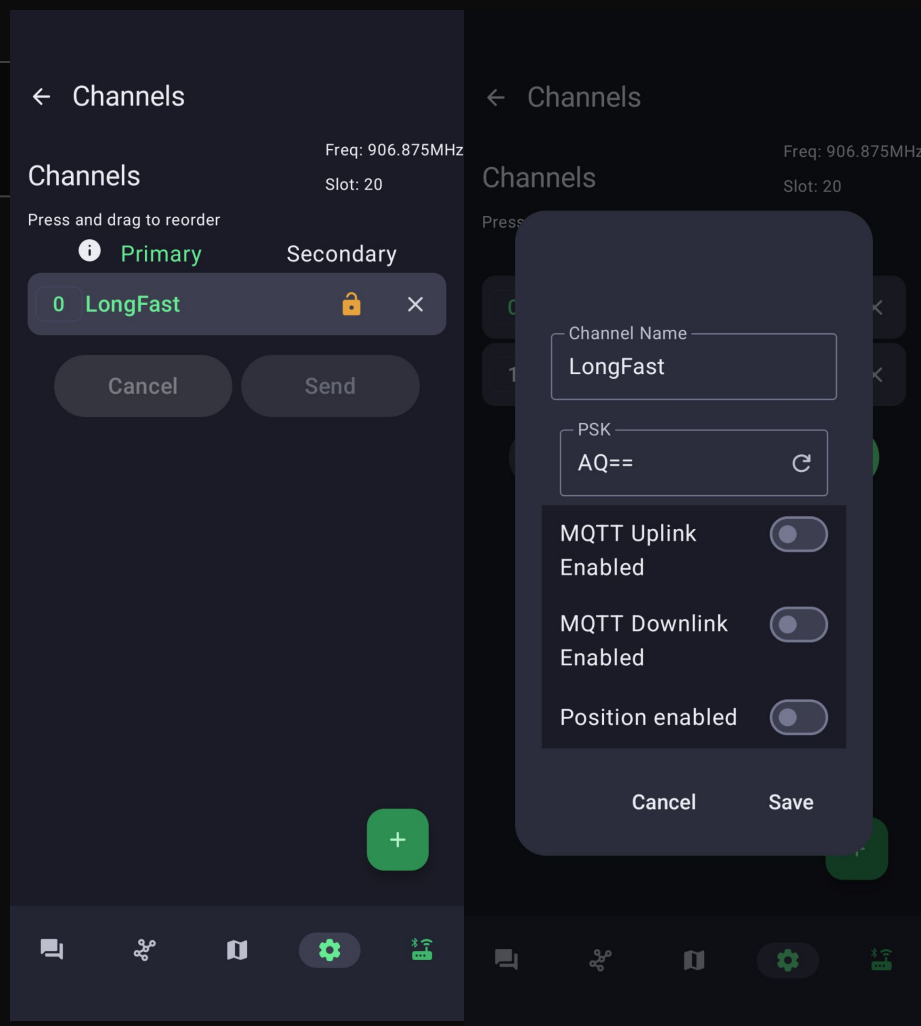
Channels



- Settings > Channels
 - Key auto-generated based on name
 - Limit 7 channels
 - LongFast is default primary channel
 - Changing Primary forces manual frequency slot configuration
 - Uplink/Downlink: allows send/receive from public internet gateway
 - Gateway configured in MQTT settings
 - Position: shares location if active

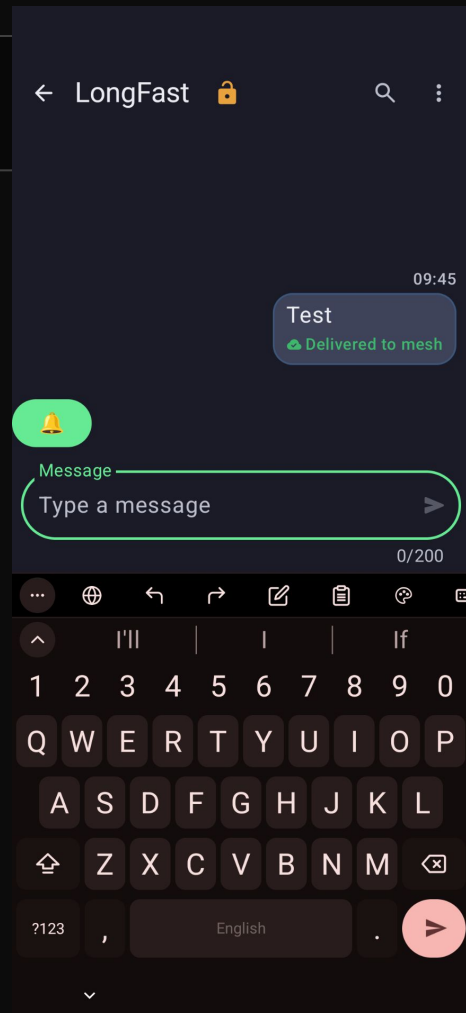
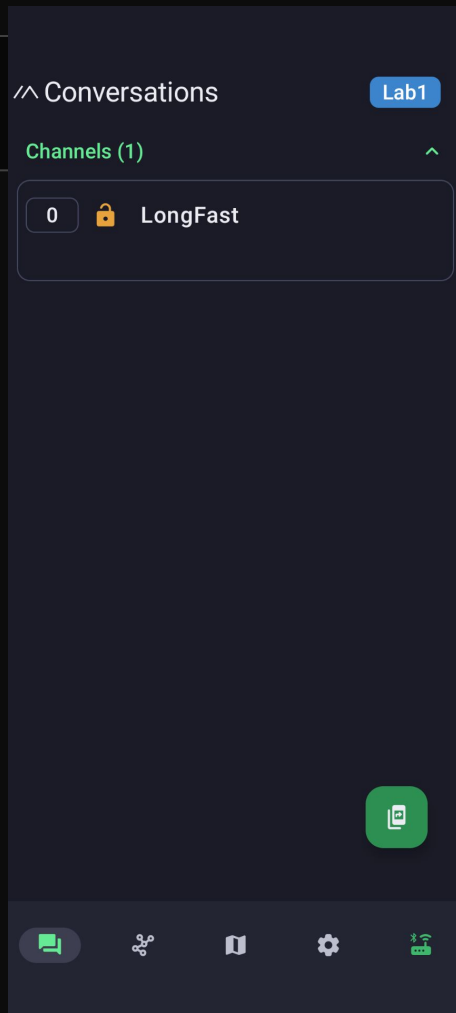
Make a Channel

- Settings > Channels
- Key auto-generated based on name
- Limit 7 channels
- LongFast is default primary channel
 - Changing Primary forces manual frequency slot configuration
- Uplink/Downlink: allows send/receive from public internet gateway
 - Gateway configured in MQTT settings
- Position: shares location if active



Messaging

- Each channel has a name and a pre-shared key (AES-256-CTR symmetric encryption)
- Default channel has name of frequency preset settings and empty password (base64-encoded as AQ==)
- Can also message nodes directly once you've seen them (Public Key Cryptography, currently X25519 Diffie-Hellman elliptic curve key exchange (rather than RSA) for post-quantum readiness)



Messaging



Unknown



Received

Direct message received and acknowledged



Queued

Waiting for your node to connect



Enroute

Waiting for a node receive/ack it



Delivered

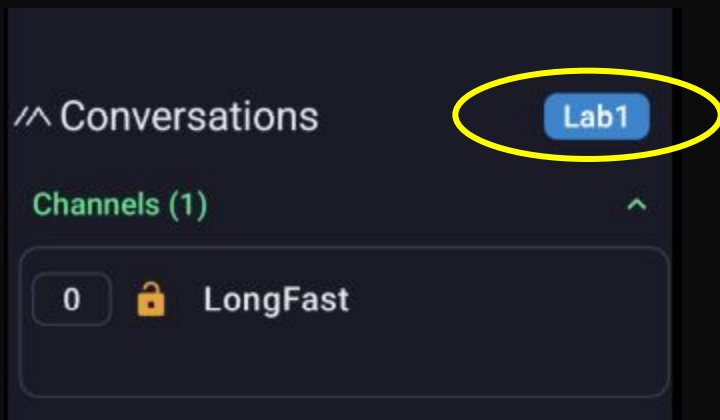
Message received by mesh



Error

No network or hop/rebroadcast limit exceeded

Viewing Data



Details
GFX-WisLab-01

Details



RAK WisBlock 4631
(rak4631)

Supported

Short Name

Wis1

Device Role

CLIENT

Node ID

!3fcd20c8

Node Number

1070407880

Last heard

3 min. ago

User ID

!3fcd20c8

Uptime

6m 6s

Public Key

8yuV0bwi1Q7RR4WtrtMJS4Y85YXYEfyj8Kg
FAW/TqXc=

I want one

Details
GFX-WisLab-01

Signal Quality

Device Metrics

Environment
Metrics

Temp
76°F

Hum
50%

Dew Point
57°F

Pressure
984 hPa

Gas Resistance
47 MΩ

IAQ
50

Air-Quality Metrics

Power Metrics

Host Metrics



Share and Add Channels

Device camera adds channel info to app. Historically buggy on iOS app.

// Conversations

Lab1

Channels (1)

0



LongFast



Scan Shared Channels NFC



Scan Shared Channels QR Code



Input Shared Channels URL



Share Channels QR Code



Conclusion

QUESTIONS?

thank you!

you have to show us what you build (it's the law)

dani@societyofhackers.com

greyfox@wcassembly.com

<https://github.com/witchofthewires/protomesh>

<https://guihackersguildpgh.com>